TEACHER EDITION: Virtual Lab Pre-Lab, Empirical Formula of Magnesium Oxide

- **Students write everything on this sheet;** the simulation has no data tables, typed answers or score. **Every student gets different numbers** (crucible + lid 25.000 to 35.000 g, Mg 0.100 to 0.200 g, a scatter within about 1%), so check the math, not the numbers. The grey answers below use **sample data**, not any one student's.
- **Check in this order:** Box 1 + Box 2 = Box 3; Box 5 = Box 4 − Box 3; mol Mg = Box 2 ÷ 24.3; atoms = mol × 6.02 × 10^{23}; mol O = Box 5 ÷ 16.0; O : Mg ≈ 1 : 1 (about 0.97 to 1.03), so MgO. The side product is magnesium **nitride**, Mg_3N_2, not nitrate.

Goal: Use experimental data to determine the ratio at which two or more elements combine.

Introduction: In the real lab you will react magnesium metal with the oxygen gas in the air to find the ratio at which magnesium atoms and oxygen atoms combine. First you will practice the setup and the calculations in a virtual lab. Run the simulation with this sheet beside you and **write every reading, observation and calculation here.** The simulation tells you which box to fill in.

Question 1. Write the chemical reaction you expect to happen between magnesium and oxygen gas. Balance it.

$2\ Mg + 1\ O_2 \rightarrow 2\ MgO$ (magnesium oxide, from Mg^{2+} and O^{2-})

Data table

Box	What you measure	Your reading
1	Mass of crucible + lid	28.4127 g
2	Mass of Mg	0.1523 g
3	Mass of crucible + lid + Mg	28.5650 g
4	Mass of crucible + lid + product, after the reaction	28.6649 g
5	Mass of oxygen added (Box 4 − Box 3)	0.0999 g

Observations (the flame, the magnesium, the product, what you see when you add water):

Bright white glow when the Mg burns; a white powder, sometimes with pale yellow-green bits (magnesium nitride); hissing and steam when the water goes on; gas wisps rising from the crucible (the magnifier shows NH_3; in the real lab there is also a sharp ammonia smell); a white, dry powder at the end.

Procedure (what you do in the simulation)

1) **Mass the clean, dry crucible and lid.** Drag them onto the balance. Record the reading in **Box 1**, then set them back on the ceramic pad.
2) **Mass the piece of magnesium.** Drag it onto the balance and record it in **Box 2**. Put the Mg in the crucible, then mass the crucible + lid + Mg and record it in **Box 3**. Check: Box 1 + Box 2 should equal Box 3.
3) **Set up the heating.** The ring stand, clay triangle and Bunsen burner are set up for you. Set the crucible, with its lid, in the clay triangle.
4) **Heat the magnesium with the Bunsen burner.** Turn the bottom of the burner up to the end of the scale, so the tip of the flame touches the bottom of the crucible. The lid is on with a slight crack so air can get in. Watch the magnifier and write what you see in Observations.
5) **Once the Mg stops glowing, add 6 to 7 drops of DI water.** Set the lid on the ceramic pad, hold the DI water bottle above the open crucible and add the drops one at a time. Then reheat to boil the water off. Write what you see. (In the real lab you will also notice a sharp smell.)
6) **Once the product is dry, turn the heat off and cool for 5 minutes.** The simulation speeds up the clock. Do the **While Cooling** questions.
7) **Mass the crucible + lid + product.** Record it in **Box 4**. Then calculate **Box 5**.

While Cooling

1. Calculate the number of moles of Mg used in this experiment. Show your work with units.

   0.1523 g Mg × (1 mol Mg / 24.3 g Mg) = **0.006267 mol Mg** (24.31 g/mol also accepted)

2. Calculate the number of Mg atoms that were in the ribbon.

   0.006267 mol Mg × (6.02 × 10^{23} atoms / 1 mol) = **3.773 × 10^{21} atoms Mg**

3. Do you expect the mass of the crucible + lid + product to be smaller, larger, or the same as before the reaction? Explain your answer.

Larger. The magnesium combined with oxygen from the air. The oxygen adds mass, and no mass was lost (conservation of mass).

After Cooling

1. Calculate the moles of oxygen present in the product. (Use 16 g/mol for the molar mass of oxygen, not 32 g/mol.)

0.0999 g O × (1 mol O / 16.0 g O) = **0.006244 mol O**

2. Compare the number of moles of magnesium to the moles of oxygen. What do you notice?

0.006267 mol Mg and 0.006244 mol O are almost equal: O : Mg = 0.996, about **1 : 1**.

3. Based on your data, what formula did the experiment lead to?

MgO (1 Mg : 1 O)

Look closer: magnesium and nitrogen (use the magnifier)

Why do we add water?

Air is about 78% nitrogen gas, N_2. When magnesium burns it reacts with some nitrogen as well as the oxygen, making a pale yellow-green powder called magnesium **nitride**. Any nitride left would spoil the result: the product would not be only MgO.

That is why you add water after the Mg stops glowing. The water reacts with magnesium nitride and **converts it to magnesium hydroxide**, $Mg(OH)_2$, and ammonia gas. When you reheat, the water boils off and the magnesium hydroxide changes into MgO, so everything left is MgO.

N1. Magnesium forms Mg^{2+} and nitrogen forms N^{3-}. Write the formula for magnesium nitride and explain how the charges decide the subscripts.

Mg_3N_2. 3 Mg^{2+} gives +6 and 2 N^{3-} gives −6, so the charges cancel.

N2. Balance the reaction of magnesium with nitrogen gas: _____ Mg + _____ N_2 → _____ Mg_3N_2

3 Mg + 1 N_2 → 1 Mg_3N_2

N3. Balance the reaction of the water with magnesium nitride: _____ Mg_3N_2 + _____ H_2O → _____ $Mg(OH)_2$ + _____ NH_3

1 Mg_3N_2 + 6 H_2O → 3 $Mg(OH)_2$ + 2 NH_3

N4. One product in N3 is a gas. Which one is it? (In the real lab you will notice its sharp smell.) Where does it go?

Ammonia, NH_3. It leaves the crucible as a gas (the simulation shows it rising; in the lab you smell it), so it is not part of the final mass.

N5. Balance the change that happens when you reheat: _____ $Mg(OH)_2$ → _____ MgO + _____ H_2O. In one sentence, why does the water step let us treat the final product as all MgO?

1 $Mg(OH)_2$ → 1 MgO + 1 H_2O. The water turns any Mg_3N_2 into $Mg(OH)_2$, and heating turns the $Mg(OH)_2$ into MgO, so no nitrogen is left in the product.

What if…?

For each mistake, circle what would happen to your numbers and explain in one sentence. Assume everything else is done correctly.

What goes wrong	Mass of crucible + lid + product would be:	The calculated O : Mg ratio would be:	Why?
a) The lid is on **too wide**: white smoke escapes	**too low**	**too low**	White MgO smoke leaves the crucible, so less product is weighed and less "oxygen" is found.
b) The lid is on **too tight**: no gap	**too low**	**too low**	Not enough O_2 gets in, so some Mg never burns; less mass is gained, and less oxygen is found.
c) You **skip the water** step	**too low**	**too low**	Mg_3N_2 is left. Each Mg in it holds only 2/3 of a nitrogen (about 9.34 g per mol Mg), far less than the 16.0 g of oxygen in MgO, so the mass is low and the ratio would come out near 0.58.